DIFFERENTIATION FORMULAS

DIFFERENTIAL CALCULUS

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GRADING SYSTEM

FINAL GRADE

30%

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SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Differentiation Formulas

- Constant function
- Power Rule
- Constant Multiple Rule
- Sum and Difference Rule
- Product Rule
- Quotient Rule

DIFFERENTIATION FORMULAS

DERIVATIVE OF A CONSTANT FUNCTION

$$\frac{d}{dx}(c) = 0$$

Determine the derivative of

$$y = 5$$

$$\frac{dy}{dx} = \frac{d}{dx}(5)$$

$$\boxed{\frac{dy}{dx} = 0}$$

ans

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Determine the derivative of

$$y = t^4$$

$$\frac{dy}{dt} = \frac{d}{dt}(t^4)$$

$$\frac{dy}{dt} = 4t^{4-1}$$

$$\boxed{\frac{dy}{dt} = 4t^3}$$

ans

CONSTANT MULTIPLE RULE

$$\frac{d}{dx}\big(cf(x)\big) = c\frac{d}{dx}f(x)$$

Determine the derivative of

$$y = 3x^4$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(3x^4\right)$$

$$\frac{dy}{dx} = 3\frac{d}{dx}\left(x^4\right)$$

$$\frac{dy}{dx} = 3\left(4x^{4-1}\right)$$

$$\frac{dy}{dx} = 3\left(4x^3\right)$$

$$\boxed{\frac{dy}{dx} = 12x^3}$$

ans

SUM AND DIFFERENCE RULE

$$\frac{d}{dx}\big(f(x)+g(x)\big)=\frac{d}{dx}f(x)+\frac{d}{dx}g(x)$$

$$\frac{d}{dx}\big(f(x)-g(x)\big)=\frac{d}{dx}f(x)-\frac{d}{dx}g(x)$$

Determine the derivative of

$$y=8x^7+60x^4-16x^3+30x^2-6$$

$$\frac{dy}{dx}=8(7x^6)+60(4x^3)-16(3x^2)+30(2x)$$

$$\boxed{\frac{dy}{dx}=56x^6+240x^3-48x^2+60x}$$

ans

$$\frac{d}{dx}[f(x)g(x)]$$

$$= f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$$

Determine the derivative of

$$y = 6x^3 7x^4$$

$$\frac{dy}{dx} = 6x^3\frac{d}{dx}(7x^4) + 7x^4\frac{d}{dx}(6x^3)$$

$$\frac{dy}{dx} = 6x^3(28x^3) + 7x^4(18x^2)$$

$$\boxed{\frac{dy}{dx} = 168x^3 + 126x^2}$$

ans

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$$

$$= \frac{g(x)\frac{d}{dx}f(x) - f(x)\frac{d}{dx}g(x)}{[g(x)]^2}$$

Determine the derivative of

$$y = \frac{x^2 + x - 2}{x^3 + 6}$$

$$\frac{dy}{dx} = \frac{(x^3+6)\frac{d}{dx}(x^2+x-2) - (x^2+x-2)\frac{d}{dx}(x^3+6)}{(x^3+6)^2}$$

$$\frac{dy}{dx} = \frac{(x^3+6)(2x+1) - (x^2+x-2)(3x^2)}{(x^3+6)^2}$$

$$\frac{dy}{dx} = \frac{(2x^4+x^3+12x+6) - (3x^4+3x^3-6x^2)}{(x^3+6)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{-x^4-2x^3+6x^2+12x+6}{(x^3+6)^2}}$$

ans

If $y = \frac{1}{x}$, find y'.

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{x}\right)$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(x^{-1}\right)$$

$$\frac{dy}{dx} = -x^{-2}$$

$$\boxed{y' = -\frac{1}{x^2}}$$

ans

Differentiate the function

$$y = \sqrt{x}\,(x-1)$$

$$\frac{dy}{dx} = (x-1)\frac{d}{dx}\left(\sqrt{x}\right) + \sqrt{x}\frac{d}{dx}(x-1)$$

$$\frac{dy}{dx} = (x-1)\frac{d}{dx}\left(x^{1/2}\right) + \sqrt{x}\frac{d}{dx}(x-1)$$

$$\frac{dy}{dx} = (x-1)\left(\frac{1}{2}x^{-1/2}\right) + \sqrt{x}(1)$$

$$\frac{dy}{dx} = \frac{(x-1)}{2\sqrt{x}} + \sqrt{x}$$

$$\frac{dy}{dx} = \frac{(x-1)}{2\sqrt{x}} + \sqrt{x}\cdot\frac{2\sqrt{x}}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{(x-1)+2x}{2\sqrt{x}}$$

$$\boxed{\frac{dy}{dx} = \frac{3x-1}{2\sqrt{x}}}$$

ans

Find the **slope** of the tangent line to the curve

$y = \frac{\sqrt{x}}{1+x^2}$ at the point $\left(1, \frac{1}{2}\right)$.

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\sqrt{x}}{1+x^2}\right)$$

$$\frac{dy}{dx} = \frac{(1+x^2)\frac{d}{dx}(\sqrt{x}) - (\sqrt{x})\frac{d}{dx}(1+x^2)}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{(1+x^2)\frac{d}{dx}(x^{1/2}) - (\sqrt{x})\frac{d}{dx}(1+x^2)}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{(1+x^2)\left(\frac{1}{2\sqrt{x}}\right) - (\sqrt{x})(2x)}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{(1+x^2)\left(\frac{1}{2\sqrt{x}}\right) - (\sqrt{x})(2x)\cdot\frac{2\sqrt{x}}{2\sqrt{x}}}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{\frac{(1+x^2)-4x^2}{2\sqrt{x}}}{(1+x^2)^2}$$

$$\frac{dy}{dx} = \frac{(1+x^2)-4x^2}{2\sqrt{x}(1+x^2)^2}$$

Find the **slope** of the tangent line to the curve

$y = \frac{\sqrt{x}}{1+x^2}$ at the point $\left(1, \frac{1}{2}\right)$.

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{(1+x^2) - 4x^2}{2\sqrt{x}(1+x^2)^2}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{(1+1^2) - 4(1^2)}{2\sqrt{1}(1+1^2)^2}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{2-4}{2(2)^2}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \frac{-2}{8}$$

$$\boxed{\left.\frac{dy}{dx}\right|_{x=1} = -\frac{1}{4}}$$

ans

SEATWORK